



Full Length Article

Impact of multi-scale microstructural heterogeneities on the mechanical behavior of additively manufactured and post-processed Nb-based C103 alloy

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ABSTRACT

Laser powder-bed fusion (LPBF) processed Nb-based alloy C103 (Nb-10Hf-1Ti wt.%) develops a complex, hierarchical microstructure comprising a fine-scale solidification cell structure, overlaid with a dense dislocation-network outlining the cell boundaries, within the primary grains. Additionally, sub-grain boundaries and a fine-scale dispersion of nano-sized hafnium oxide precipitates, possibly forming during solidification, decorate the solidification cell boundaries as well as exist within the cells. This complex hierarchical microstructure results in impressive tensile mechanical properties. Post-build stress-relieving annealing and hot isostatic pressing (HIP) largely annihilates the solidification cell structure and associated dislocation network, lowering the strength but with substantial recovery of tensile ductility. Nevertheless, the resulting microstructure offers higher strengths as compared to their wrought counterparts.

1. Introduction

Fusion-based additive manufacturing (AM) has opened new opportunities for fabricating refractory alloys into complex parts [1–4]. Refractory alloys are attractive because of their high creep strengths at high temperatures [1,5,6]. The recommended operating temperature range for commercially available refractory alloys as structural materials for space propulsion systems are significantly higher compared to superalloys, stainless steels, and ODS steels [7]. Niobium (Nb) is a desirable candidate from the refractory metal alloy family given its low ductile-to-brittle transition temperature of < 142 K [8] making it more amenable to various forming processes, and its low density of 8.57 g/cc compared to other refractory alloys [8]. A Nb-based refractory alloy of particular interest is Nb-10Hf-1Ti in wt%, commercially referred to as C103. The Nb-based alloy has a BCC single-phase solid solution microstructure and a bulk density of 8.85 g/cc [8]. Hf and Ti impart significant solid solution strengthening to Nb without compromising on ductility [9,10]. The strength-ductility combination in C103 makes it amenable to various fabrication routes (welding, forming, etc.) [11]. Since C-103 has been used in aerospace applications, such as rocket nozzles, since the

days of the Apollo missions, and is an important candidate alloy for future hypersonic applications [12–14], AM-based printing of near-net shape components is of great interest. However, there are only a limited number of studies on AM processed C103. Despite that, various AM techniques have been employed to print C103 such as directed energy deposition [15], electron beam powder bed fusion [16], and laser powder bed fusion (LPBF) [1,12,17].

LPBF is the technique of choice owing to the advantages it offers, e. g., better surface finish and precision, finer microstructural features, etc. [18]. Furthermore, the complex thermal history and high cooling rates in LPBF result in non-equilibrium, heterogeneous microstructures including high dislocation density, and cellular structures in some systems [19–27]. Such microstructures have a direct consequence on the structural properties of metallic alloys and are yet to be understood. Mireles *et al.* [12,28] was able to successfully fabricate C103 alloy using LPBF and observed tensile yield strengths of 540 MPa. Awasthi *et al.* [17] also printed C103 alloy using LPBF and stress-relieved it, and consequently observed a room temperature tensile yield strength of 410 MPa. The increase in strength observed in the study, compared to wrought C103, was attributed to the presence of dislocation cell networks and

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low angle boundaries in the alloy, and no second phase precipitates were observed [17]. While both the studies successfully printed C103 using LPBF exhibiting yield strengths much higher compared to conventionally processed C103, there is still little insight on the impact of the hierarchical microstructure, arising from the LPBF process, along with the underlying complex dislocation structure, on the mechanical properties of C103.

Also, while LPBF dramatically enhances the strength, the high residual stresses can result in early failure, necessitating a stress-relieving treatment. Additional defects are porosities (even in parts with > 99% relative density) and columnar grains [20,25,29]. In such a scenario, an additional step of hot isostatic pressing (HIP) appears promising in alleviating the properties making the printed part denser and more isotropic [15,16,28]. While the previous studies on AM processed C103 have reported very good mechanical properties, they lack a detailed analysis of the underlying hierarchical microstructural features leading to these properties. Additionally, the influence of post-processing treatments, such as stress-relieving as well as HIP'ing, on the hierarchical microstructure and its consequent impact on the mechanical properties has not been addressed. These aspects form the basis of the present study on LPBF processed C103.

2. Material and methods

Spherical pre-alloyed C103 powders acquired from H.C Starck, were analyzed using the Microtrac S3500 laser diffraction analyzer for

powder size distribution (PSD) analysis. The D_{90} was found to be 48 μm . 0.042 wt% O concentration was estimated in the powder using the LECO ON736 infrared analyzer. Samples were printed on the Trumpf TruPrint 1000 equipped with a 55 μm laser beam diameter and layer height of 50 μm . Bi-directional scanning strategy was employed with 90° scan rotation every layer. The prints were carried out in an inert Ar atmosphere, with the chamber oxygen level maintained below 100 ppm. The as-built block (AB) was subjected to two different post-deposition processing treatments. For the first treatment, AB was stress relieved under vacuum ($\sim 10^{-5}$ torr) at 1100 °C for 1 h and furnace cooled (referred to as AB+SR). In the second treatment, the AB block was HIP'ed at 1300 °C under 206 MPa for 4 h (referred to as AB + HIP). The stress-relieving parameter has been adopted from literature [15,28]. The densities for all the conditions were evaluated based on the Archimedes' principle according to ASTM B962, and the relative densities were computed with respect to the bulk density of 8.85 g/cc.

Uniaxial tensile tests at room temperature (strain rate of 10^{-3} s^{-1}) have been carried out using Kammrath-Weiss, 5 kN tensile module on tensile specimens with 10 mm gage length, 4.7 mm gage width, and 1 mm thickness. The samples were machined perpendicular to the build direction using a KENT USA (WSI-200) wire-type electrical discharge machine. Microstructural characterization was performed using an FEI Nova NanoSEM 230, and Electron Back-Scattered Diffraction (EBSD) using an Thermo Fisher Apreo 2C SEM equipped with Ametek EBSD detector and Energy Dispersive Spectroscopy (EDS) detectors. EBSD datasets were analyzed using TSL OIM Analysis™ ver8. Microstructural

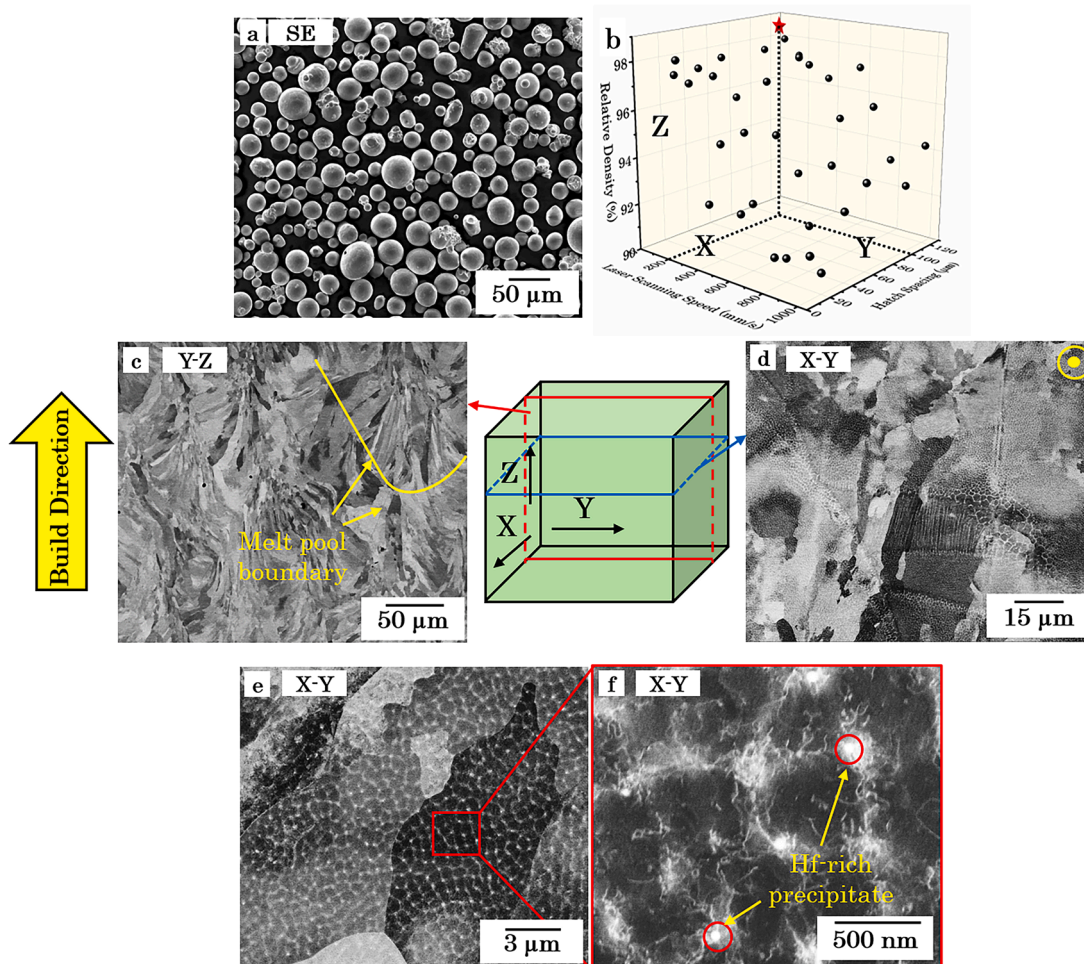


Fig. 1. (a) SE image of the starting powders. (b) 3D plot of relative density vs. the laser scanning speeds and hatch spacing, the best parameter is indicated by the red star. (c) and (d) are BSE images of YZ and XY planes of AB condition respectively. (e) Image of the region of interest from which a higher magnification image of the dislocation cell network is shown in (f).

quantification using micrographs acquired thus has been carried out using the MIPAR image analysis software [30]. A dual-beam focused ion beam milling system (FEI Nova 200 Nanolab FIB/SEM) was utilized for the preparation of a thin foil for transmission electron microscopy (TEM). TEM for the thin foil was carried out in FEI Tecnai G2 F20 ST S/TEM operating at 200 kV, and equipped with an EDS detector (EDAX LLC, Mahwah, New Jersey).

3. Results and discussion

The morphology of the pre-alloyed C103 powders is shown in Fig. 1 (a). The powder particles are predominantly spherical. The laser power, scanning speed, and hatch spacing used for printing this alloy were optimized for a constant layer thickness of 50 μm . As seen in the 3D scatter plot in Fig. 1(b), the highest relative density of $\sim 99\%$ was achieved at 180 W laser power, 200 mm/s scan speed, and 100 μm hatch spacing. The as-built (AB) block was further analyzed along the build direction (YZ plane) and the transverse direction (XY plane), as shown in the backscatter SEM images in Figs. 1(c) and (d) respectively. The melt pool boundaries are evident along the YZ plane in Fig. 1(c), (a representative boundary marked in yellow), and elongated grains are also observed. Therefore, the starting AB condition is characterized by some porosities and elongated grains which can be detrimental to mechanical properties resulting in early failure and anisotropic properties. The higher magnification backscatter SEM image of the XY plane in Fig. 1(d) reveals a cellular solidification microstructure within the grains, with bright intercellular channels (presumably rich in Hf) surrounding cells exhibiting a darker contrast. Scanning transmission electron microscopy (STEM) - energy dispersive spectroscopy (EDS) mapping presented in the Supplementary Section, S1 confirms some Hf enrichment in the

intercellular channels. Higher magnification backscatter SEM micrographs in Figs. 1(e) and (f) are from a strongly diffracting grain with a high channeling contrast, revealing a dislocation cell pattern overlapping with the solidification cell structure. Fine-scale (~ 50 nm), bright particles were observed in the AB condition which have been found to be rich in both Hf and O based on STEM-EDS mapping. The corresponding STEM-EDS maps are provided in the Supplementary Section, S2. The precipitates are HfO_2 that have been observed in C103 through various studies [12,15,31–33]. In the present study, EBSD and XRD data provided in Supplementary Section, S3 confirm the presence of monoclinic- HfO_2 in the LPBF processed and stress relief annealed C103. While the exact composition of these precipitates cannot be determined based on these results, it should be noted that the monoclinic- HfO_2 phase has a limited range of off-stoichiometry in the Hf-O binary phase diagram [34]. Therefore, it can be assumed that the composition of the Hf-rich precipitates are HfO_2 .

As previously stated, the AB C103 samples were treated to two distinct post-processing routes, i.e., stress-relieving treatment (referred to as AB + SR), and hot isostatic pressing (AB + HIP). The XY plane images of both AB + SR and AB + HIP are shown in Figs. 2(a) and (c) respectively. In comparison to AB, both conditions exhibited coarser Hf-oxide rich precipitates at the high angle grain boundaries (HAGBs), albeit of different size distributions, as revealed by the insets in Figs. 2(a) and (c). These images also show precipitates at the low angle boundaries besides HAGBs, as indicated by yellow arrows in the inset in Fig. 2(c). The size distributions and area fractions of these precipitates are provided in Table 1. Unlike AB + SR, there are pockets of recrystallized grains ($\sim 10 \pm 1$ μm grain size) in AB + HIP which is confirmed by EBSD data from the AB + HIP condition provided in Supplementary Section, S5. The absence of re-crystallization in AB +

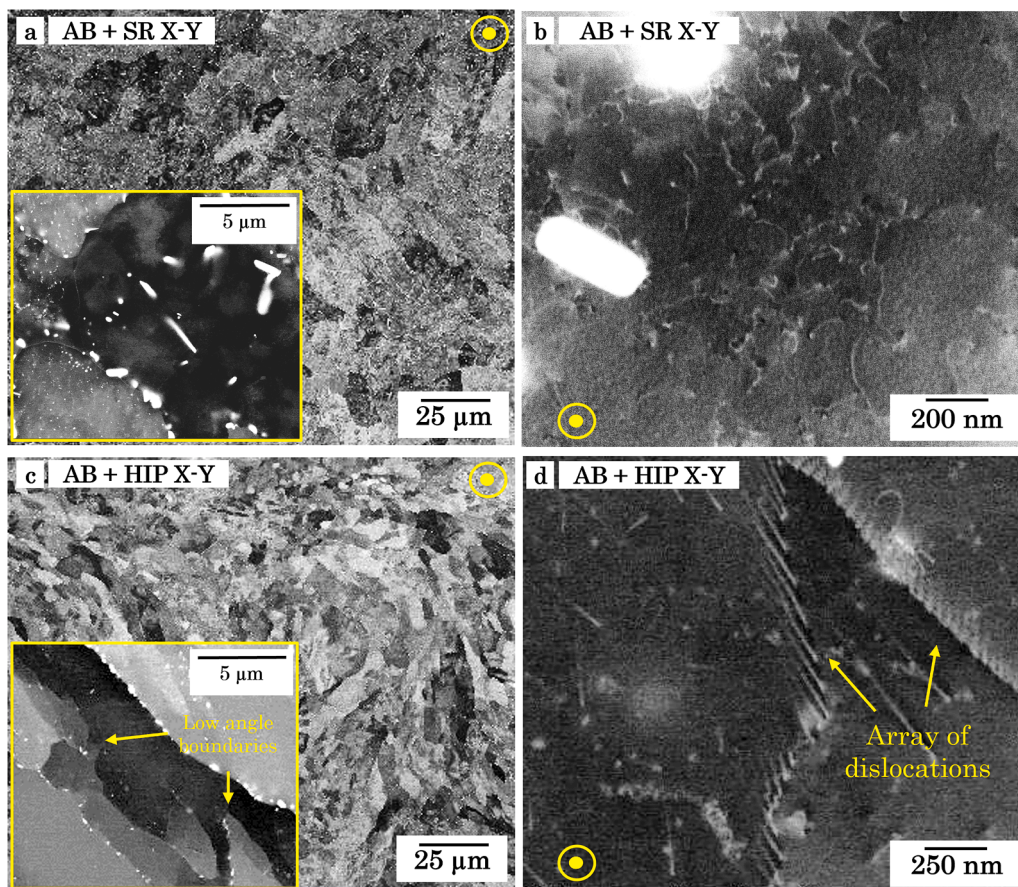


Fig. 2. (a) and (c) Low magnification images of AB + SR and AB + HIP respectively with insets showing the region of interest for higher magnification images of the dislocation structure in (b) for AB + SR, and (d) for AB + HIP.

Table 1

Size distribution and area fraction of Hf oxide rich precipitates in AB, AB + SR, and AB + HIP C103 conditions.

Hf oxide rich Precipitate	As-built	As-built + Stress-relieved	As-built + HIP
Average Size	54 ± 20 nm	107 ± 80 nm	100 ± 63 nm
Distribution			
Area Fraction	1.0 ± 0.3%	3.8 ± 1.0%	2.1 ± 0.3%

SR from the stress-relieving treatment is consistent with the observations by Mireles *et al.* [28] on LPBF C103. Higher magnification images of AB + SR and AB + HIP conditions, shown in Figs. 2(b) and (d), reveal that the solidification cell structure and dislocation cell network observed in the AB condition (Figs. 1(e) and (f)) are no longer present. Thus, the system has largely undergone recovery from the post-processing treatments. Additionally, the underlying dislocation structure in the respective conditions are also revealed in the same figures. Low angle boundaries due to polygonization during the recovery process, are also present in Fig. 2(d).

Large area EBSD mapping of the initial AB condition (provided in the **Supplementary Section, S4**) establishes the absence of any strong texture in these builds. Figs. 3(a) and (d) show the inverse pole figures of AB and AB + SR respectively. The KAM maps in Figs. 3(b) and (e) for AB and AB + SR respectively show that the local misorientation in AB + SR is quite low compared to AB which further substantiates the recovery after stress-relieving. The EBSD data for AB + HIP condition is provided in the **Supplementary Section, S5** which confirms the presence of pockets of recrystallized grains. The fraction of misorientation boundaries <15° was further categorized into LABs (Low angle boundaries, 1°–5°) and MABs (Medium Angle Boundaries, 5°–10°) for all three conditions, as shown in Table 2. The representative grain boundary maps are also provided in the **Supplementary Section, S6**.

From the EBSD analysis, the increase in average grain size (based on high angle grain boundaries or HAGBs) going from AB to AB + SR and AB + HIP is very minimal with the grain sizes across all three falling in the range of 45–55 µm. This can be attributed to the Hf-oxide rich particles pinning grain boundaries and preventing grain growth. The misorientation profiles for AB and AB + SR are shown in Figs. 3(c) and (f) respectively. In Fig. 3(c), it is evident that there are distinct LABs in the AB condition in addition to the solidification cell structure. The misorientation across the LAB is ~ 2.5°, whereas the misorientation across the dislocation cell wall at the intercellular region is even lower (< 0.5°) and could not be accurately ascertained from conventional EBSD measurements employed in the present study. This highlights that the dislocation cell pattern in AB C103 is different from sub-grains accommodated by LABs which typically correspond to higher misorientations [35]. The misorientation profile across a LAB in AB + SR, indicated by the yellow arrow as seen in Fig. 3(f) also shows a similar misorientation. Thus, the LPBF printed AB condition of C103 demonstrates a hierarchical microstructure at various length scales, comprising a cellular solidification structure overlaid with dislocation cell walls, fine scale Hf-oxide nanoprecipitates decorating the cell walls as well as within the solidification cells, LABs and HAGBs.

It is interesting to note that the dislocation cell network seen here has been observed primarily in LPBF printed FCC-based alloys such as austenitic stainless steels and Ni-base superalloys [23,24,36–39]. However, reports of such a network in LPBF printed BCC-based alloys are quite limited. Recently, Liu *et al.* [40] observed similar features in an LPBF printed TiNbTaZrMo RCCA. Tekumalla *et al.* [41] have attributed such a dislocation cell network to the moduli/yield-strength differences arising from the heterogeneities resulting from LPBF printing, and the subsequent cyclic heating/cooling they undergo as the printing proceeds in a multi-layered fashion.

The representative engineering stress-strain curves for all the three conditions of LPBF printed C103 investigated in this study are shown in

Fig. 4(a). The average values of yield stress (YS) and ultimate tensile stress (UTS) are included in Table 3, revealing a distinct difference between the three conditions, with AB C103 demonstrating both high YS and UTS, while it is the lowest in AB + SR. The area fraction of Hf-oxide is highest in AB + SR, as seen in Table 1, and the size scale of these precipitates is comparable in case of AB + SR and AB + HIP conditions. Therefore, despite the comparable size of the precipitates in these two conditions, the higher area fraction of the oxides in case of AB + SR, markedly decreases the YS presumably due to lower solid solution strengthening from Hf coupled with coarsening of the Hf-oxide precipitates. From Table 2, it is evident that the fraction of LABs and MABs in the AB condition is comparable to the AB + HIP condition, however the fraction of LABs is significantly high in case of the AB + SR condition. As there is a higher fraction of boundaries with larger misorientations in the first two conditions, the boundary strengthening can also be expected to be higher.

The comparable fraction of LABs, MABs, and HAGBs in AB and AB + HIP conditions of C103 can be attributed to the fact that during both LPBF processing and subsequent HIPing, the material experiences stress/plastic deformation and competing relaxation effects such as recovery and recrystallization due to the nature of the two processing routes. In the case of additive manufacturing processes such as LPBF, the rapid heating and cooling cycles ~ 10⁴ – 10⁵ K/s [25] typically induce significant thermal gradients [44–47]. As a result of the thermal cycling effect, the material typically experiences thermal stresses sufficient for plastic deformation [44–47]. However, the very same thermal cycling effect also mimics in-situ localized annealing treatment due to the repeated heating cycles [44]. Therefore, the resulting substructure in an as-built component is a result of this competitive process between thermal stress induced deformation and relaxation processes during printing. Similarly, in the case of HIPing, a material may undergo plastic deformation due to the high pressures/isostatic stresses, along with competing relaxation processes resulting from the high temperatures involved [48]. Furthermore, because of inhomogeneous strain distribution in AB C103 from printing and varying grain sizes, it is expected that recrystallization would not take place uniformly throughout the microstructure [49] in the HIP treatment of 1300 °C for 4 h at 206 MPa employed in the current study on AB C103. On the other hand, in the case of AB + SR C103, the starting AB C103 material only experiences stress-relieving processes due to the annealing treatment performed at 1100 °C for 1 h. The manifestation of this stress-relieving process is that the high dislocation density remnant from the AB condition, rearranges to form low angle misorientation boundaries within the primary columnar grain boundaries (HAGBs). This results in the higher fraction of LABs formed in case of the AB+SR condition. Lastly, but probably most importantly, it appears that the presence of the dense dislocation cell-type network contributes to the high YS in the AB condition which sets it apart from AB + HIP condition despite having similar fraction of LABs, MABs, and HAGBs. Such an increase in YS due to the presence of dislocation cell network has been reported previously in LPBF printed FCC alloys [36,38,50].

The strain hardening rates (SHR) were calculated for all three conditions and are shown in Fig. 4(b). Based on these curves, across all the three conditions, it appears that dislocation mediated plastic deformation is the only deformation mechanism, in agreement with previous reports on C103 [16,17,28]. The fully deformed microstructures were also analyzed, and the results are presented in **Supplementary Section, S7**. The deformed microstructures are characterized by extensive dislocation entanglement in the case of AB C103 as shown in S6 (a), and bands of high dislocation density walls with significant dislocation activity between the walls in the case of AB + SR C103 in S6 (b). There have been several studies carried out on the deformation and resulting dislocation structure in the case of pure Nb with varying amounts of interstitials [51–54]. It has been shown that the cell structure formation is possible in the case of elemental Nb with higher concentrations of impurities (and second phase precipitates) at room temperature early in

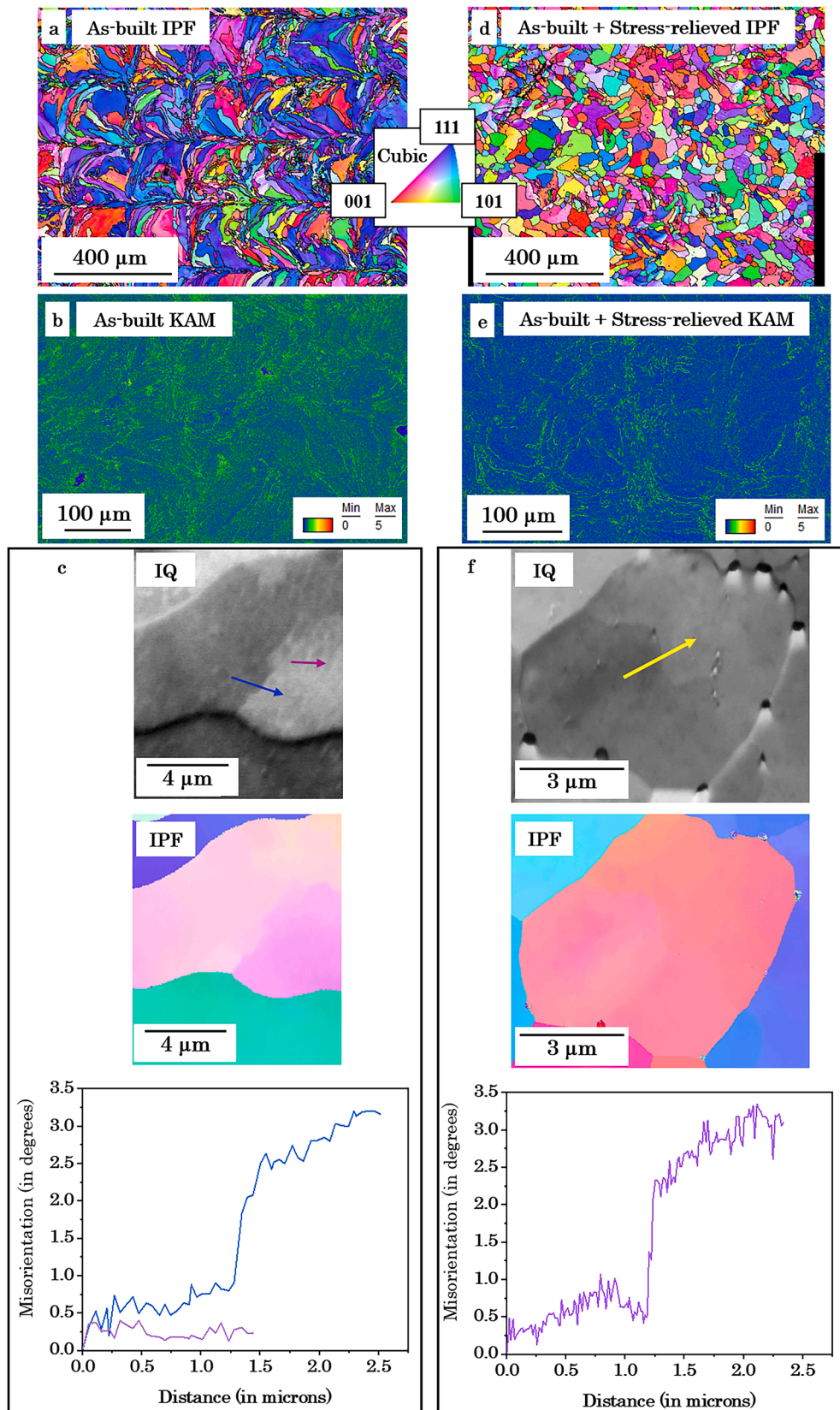


Fig. 3. (a) Inverse pole figure (IPF), (b) Kernel average misorientation (KAM) map, and (c) misorientation profile vs. distance along with representative vectors indicating the regions of interest in AB. The corresponding IPF map to the IQ containing the vectors is also shown. Similarly, the IPF, KAM, and misorientation profiles are shown in (d), (e), and (f) for AB + SR respectively. All the maps shown are from XY plane.

Table 2

Fraction of boundaries for various ranges of misorientations in AB, AB + SR, and AB + HIP C103.

	As-built	As-built + Stress-relieved	As-built + HIP
Fraction of 1°–5° Boundaries (<i>low angle, LABs</i>)	0.46	0.77	0.45
Fraction of 5°–15° Boundaries (<i>medium angle, MABs</i>)	0.11	0.08	0.13
Fraction of > 15° Boundaries (<i>high angle</i>)	0.43	0.15	0.42

the deformation process [53]. With increasing deformation, additional dislocations created were either forced or attracted into the cell walls, and further deformation led to extensive entanglements within the walls with walls becoming thicker during the process [51–53]. The tendency for Nb to form cell structures due to deformation and to showcase dislocation tangles has been attributed to cross-slip mechanism in Nb, given its intermediate to high stacking fault energy [51–53]. Furthermore, the presence of second phase precipitates such as oxides and carbides also act as sources of dislocation multiplication during deformation [53]. The dislocation structures observed in the Nb-base C103 alloy employed in the present study appear to be similar to that of pure Nb subjected to varying degrees of deformation across multiple studies [51–54]. However, further investigation needs to be carried out on the deformation behavior of the LPBF processed C103 alloy. Additionally, the AB + HIP C103 condition also exhibits the highest strain hardening followed by the AB + SR condition, and then the AB condition. Comparison with literature values for both YS and UTS of LPBF processed C103 with or without post-processing [1,17,28,42,43], as shown in Fig. 4(c) and (d) respectively, indicate that the values observed in this

study for all three conditions of C103 are close to literature values. Furthermore, the LPBF and LPBF + HIP'd C103 conditions across all studies demonstrate much higher tensile strengths compared to wrought counterpart, and C103 fabricated through solid-state powder metallurgical routes [42,43]. The presence of a dislocation cell network with overlapping solidification cell structure observed in the current study, similar to what has been reported by other studies in the case of refractory complex concentrated alloy (RCCAs) [40,55], has a significant influence on increased strengthening. This mechanism is coupled with the strengthening effects from Hf and Ti in the solid solution and nanoscale HfO₂ precipitates in the LPBF processed alloy.

4. Conclusion

In summary, we report a multi-scale heterogeneous microstructure resulting from LPBF printing of the Nb-based alloy, C103. Cellular solidification leads to the formation of Hf-rich inter-cellular channels with Hf-oxide formation within these channels. A dense dislocation cell network was found overlapping with this cellular solidification structure which dramatically enhances the tensile yield strength. Its contribution

Table 3

Average room temperature tensile properties of AB, AB + SR, and AB + HIP C103.

	As-built	As-built + Stress-relieved	As-built + HIP	Wrought C103 [42]
YS (in MPa)	486 ± 30	292 ± 7	371 ± 4	342
UTS (in MPa)	564 ± 17	416 ± 8	533 ± 3	481
ϵ_f (in%)	7 ± 3%	21 ± 6%	39 ± 1%	34%

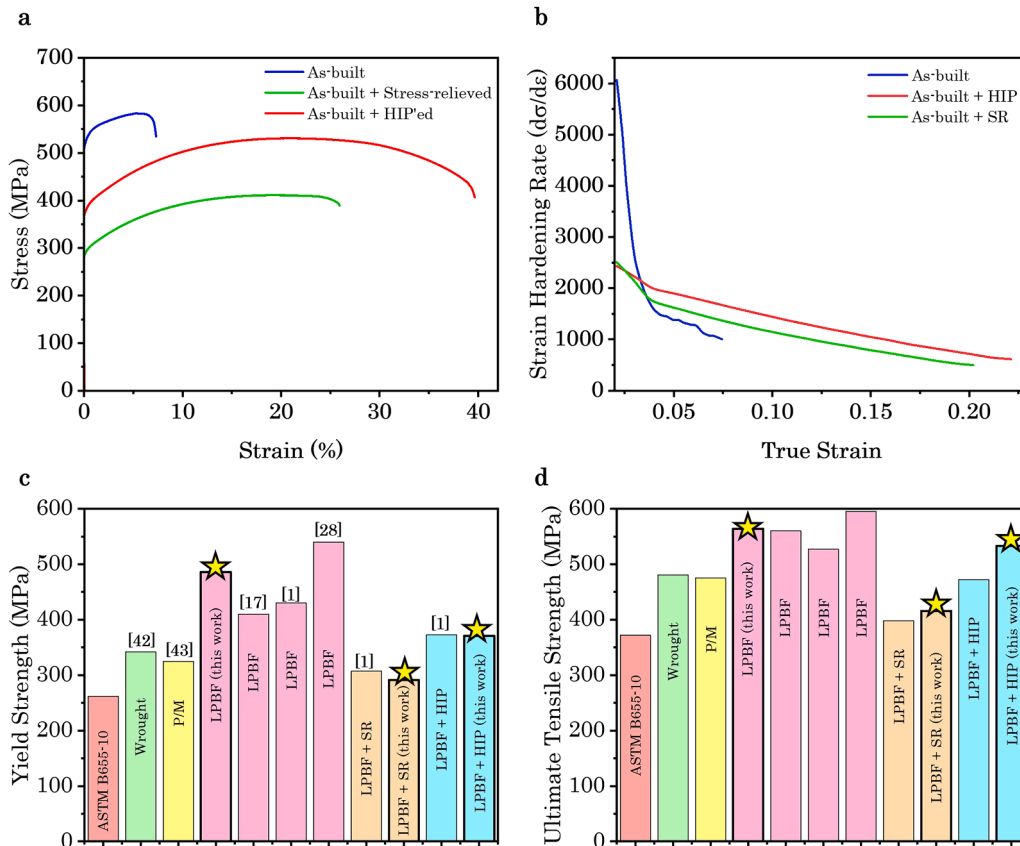


Fig. 4. (a) Representative engineering stress-strain curves and (b) strain hardening rate vs. true strain curves for all three conditions. Comparison of the tensile strength observed in AB, AB + SR, and AB + HIP with current literature values [1,17,28,42,43] in (c) and (d).

is evident when compared with AB+SR and AB+HIP conditions which are devoid of this dislocation cell network. While the post-deposition processing wipes out the dislocation cell-network, resulting in loss of strength, the ductility is substantially enhanced (AB+HIP~40% and AB+SR~21% compared to AB~7%). Therefore, the presence of Hf-oxide nanoprecipitates along with the dislocation cell network and intercellular segregation offers a microstructural template for enhanced strengthening in comparison with conventionally processed C103. These insights will help in future design of multi-scale hierarchical microstructures in AM processed Nb-base and other refractory alloys to achieve favorable combinations of room temperature tensile strength and ductility.

Data Availability Statement: The authors can provide raw data upon reasonable request.

CRedit authorship contribution statement

Advika Chesetti: Data curation, Formal analysis, Investigation, Methodology, Writing – original draft, Writing – review & editing. **Tirthesh Ingale:** Data curation, Formal analysis, Investigation, Methodology, Writing – review & editing. **Sucharita Banerjee:** Data curation, Formal analysis, Investigation. **Madhavan Radhakrishnan:** Data curation, Formal analysis, Investigation. **Narendra B. Dahotre:** Conceptualization, Funding acquisition, Project administration, Resources. **Abhishek Sharma:** Data curation, Formal analysis, Investigation, Supervision, Writing – original draft, Writing – review & editing. **Rajarshi Banerjee:** Conceptualization, Data curation, Formal analysis, Funding acquisition, Methodology, Project administration, Resources, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.mtl.2024.102230](https://doi.org/10.1016/j.mtl.2024.102230).

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